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Explainable Multi-Model Approach for Uncovering Key Predictors of Stroke-risk from ECG, EMG, Blood Pressure, and Respiratory Signals

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Abstract

Accurate and timely stroke-risk prediction is necessary to help patients at risk take guided measures, as stroke remains a leading cause of death and long-term disability worldwide. While there are several developed stroke-risk prediction models using various bio-signals, it remains unclear which signal or signal feature carries the most information towards stroke. Additionally, respiratory signals have not been included in these models, despite research showing their relation to stroke disease. We address these gaps by developing an explainable multi-modal stroke-risk prediction model that integrates respiratory signals (carbon dioxide (CO₂), and respiration flow) alongside blood pressure (BP), electrocardiogram (ECG) and electromyography (EMG). We developed a single perceptron model which achieved a prediction accuracy of 84.91% on a dataset of 64 subjects, outperforming state-of-the-art machine learning (ML) and deep learning (DL) methods. Explainable Artificial Intelligence (XAI) techniques, namely LIME, SHAP, and Anchors, were applied to interpret the model's decisions. The Model Explanation Metric Consensus (MEMC) XAI evaluation metric revealed SHAP as the most reliable explainer, which identified CO₂-derived features as critical predictors. The findings demonstrate the overlooked value of respiratory signals in stroke-risk prediction, as well as the importance of explainable multi-modal approaches in advancing stroke-risk prediction, and enabling clinicians to better understand and trust AI models for improved patient care.

Keywords: Ischemic stroke, Respiratory features, Multi-modal approach, Explainable artificial intelligence, Explainability evaluation

1 Introduction

A stroke is a clinically defined syndrome of rapidly developing symptoms or signs of focal loss of cerebral function with no apparent cause other than that of vascular origin [1]. The five common signs and symptoms of stroke are sudden numbness of the face, arm, or leg; sudden confusion or trouble speaking; sudden trouble seeing in one or both eyes; sudden dizziness or loss of coordination; and sudden severe headache with no known cause [2]. These symptoms usually last for more than 24 hours and sometimes lead to death. Stroke is a heterogeneous syndrome arising from multiple risk factors and mechanisms, with hypertension being the most important modifiable risk factor. Ischemic strokes make up 87% of all strokes [3] and occur when cerebral blood flow is interrupted due to vessel occlusion, most commonly caused by small vessel arteriosclerosis, cardioembolism, or large artery atherothromboembolism. In younger patients, ischemic stroke may result from alternative mechanisms such as extracranial arterial dissection. Hemorrhagic strokes are less common and result from vessel rupture and intracerebral bleeding [4]. According to the World Health Organization (WHO), 15 million people worldwide suffer a stroke annually, where 5 million of those die and 5 million become permanently disabled [5]. As a result, stroke is one of the leading causes of serious long-term disability as well as death. However, researchers found that stroke patients receiving early rehabilitation had a 14% lower mortality rate within 90 days after stroke [6], stressing the importance of timely intervention.

Artificial Intelligence (AI) has become widely used in healthcare [7], specifically its role in disease prediction and diagnosis [8]. Since then, stroke-risk prediction models were developed that help stroke patients and their caregivers take preventive measures. Models relying on images, such as computed tomography (CT) or magnetic resonance imaging (MRI) [9–11], are expensive and require heavy machinery. Therefore, researchers developed stroke-risk prediction models using one-dimensional bio-signals such as electroencephalogram (EEG) [12, 13], electrocardiogram (ECG) [14], electromyography (EMG) [15], and photoplethysmography (PPG) [16].

Recently, deep learning (DL) models have been employed on these bio-signals for stroke-risk prediction. Recurrent Neural Networks (RNNs) are effective DL models that capture sequential features and are thus suitable for raw time-series data [17]. Advancements in the field led to the development of more advanced versions of RNNs such as Long Short-Term Memory networks (LSTMs), Bidirectional LSTMs (BiLSTMs), Convolutional Neural Networks-LSTMs (CNN-LSTMs), and CNN-bidirectional LSTMs (CNN-BiLSTMs). LSTM models and its variants are effective for stroke-risk prediction due to their ability to capture long-term dependencies in sequential biological signals [18]. One study used 5-second windows of EEG signals of 122 seniors aged 65 and above, half of which are diagnosed with stroke [12]. The CNN-BiLSTM model achieved an accuracy of 94% in predicting stroke-risk, demonstrating the benefit of combining convolutional layers with BiLSTM for feature extraction and

temporal modeling. However, its reliance on raw EEG signals may make the model sensitive to noise and less interpretable, limiting clinical explainability and practical adoption. Another study used EMG signals with a window size of 0.1 seconds of 271 stroke patients, and 271 normal subjects, aged 65 or older [15]. The highest accuracy of 98.96% was achieved when using LSTM. The very short window size highlights LSTM's ability to capture fine-grained temporal variations, but such small windows may also be sensitive to noise, potentially affecting real-world applicability. Yu. et al. used both ECG and PPG signals simultaneously of 287 stroke patients and 287 non-stroke subjects to detect stroke with a window size of 5 seconds, and the CNN-LSTM model showed the highest prediction accuracy of 99.15% [16]. Integrating multiple physiological signals clearly improves predictive performance, showing the strength of multimodal approaches, although the increased model complexity may lead to longer training times and difficulty in real-time deployment. These results show how different studies compared the same LSTM models yet achieved different results based on the biological signal input [12, 15, 16].

Aside from LSTMs and depending on raw signals, Bouazizi and Ltifi proposed an Ensemble Echo State Network (E-ESN) for stroke-risk prediction [19]. An E-ESN contains seven individual ESNs, each having its own set of hyperparameters and reservoir configurations. The outputs from the seven ESNs are combined through an aggregation method. ESNs are a type of RNN that use a fixed, randomly connected internal reservoir to project input data into a high-dimensional space, where only the output weights are trained. The E-ESN model was tested on features of 24 participants, derived from 4-second windows of EEG signals. The model achieved an accuracy of 96.5%, proving the ability of the E-ESN in this task. However, the small number of patients used for this study is a limitation, as it reduces the generalizability of the results and increases the risk of overfitting. Another study developed an EEG feature-based machine learning model for stroke prediction [20]. EEG signals were collected during active tasks and used to classify stroke patients and healthy controls, achieving approximately 80% classification accuracy. The results emphasized the importance of low-frequency spectral features for stroke prediction, though the moderate performance suggests potential benefits from more advanced modeling or multimodal integration. Hussain and Jany utilized wearable EMG recordings from lower-limb muscles during gait to distinguish stroke patients from healthy individuals, achieving strong classification performance with an accuracy of 85.26% [21]. The analysis highlighted the relevance of EMG spectral features associated with stroke-induced gait impairments, although the reliance on gait-specific EMG recordings may limit applicability to non-ambulatory or severely impaired stroke patients, restricting the broader clinical usability of the approach.

Other studies used categorical data which consist of demographic, societal, and illness information, which uses less computational resources than signals and images. Recently, several machine learning (ML) models were developed in order to predict stroke-risk using a Kaggle dataset [22] that consists of 11 features for each subject: age, hypertension, heart disease, average glucose level, BMI, smoke, stroke, gender, work type, residence type, and marriage status [23–29]. The data contains numeric and categorical variables of 5110 subjects, where 95% of the subjects never experienced a stroke, thereby creating a data imbalance. A wide range of ML classifiers

were then compared including Random Forest, XGBoost, AdaBoost, Support Vector Classification (SVC), K-Nearest Neighbour (KNN), and ensemble methods like voting and stacking, with reported accuracies ranging from 90% to 98% [24, 26, 28, 29]. The highest reported accuracy was 98% by the voting classifier [26], which combines the predictions of multiple models and outputs the class that receives the majority vote. Ensemble voting improved prediction reliability by reducing model-specific biases, yet the heavy class imbalance in the dataset may still affect sensitivity for minority (stroke) cases, which is a key weakness in clinical applications.

While these results are promising, they remain limited by the nature of the dataset. Categorical data overlooks important physiological details and lacks temporal dynamics present in real-world patient monitoring, limiting the depth of analysis. Studies acknowledged these constraints and proposed integration with real-time data from sensors [23, 25, 27]. Moreover, the class imbalance in the Kaggle dataset [22] resulted in the generation of synthetic data [23–29], which is not always reliable. These limitations highlight the need for more robust models trained on bio-signals.

To the best of our knowledge, there is not a single study that used respiratory signals for stroke-risk prediction, despite evidence showing that risk of stroke is directly affected by the respiratory system [30–50]. Sleep-related breathing disorders are one of the major risk factors of stroke. Barnett et al. [32] found that patients with severe obstructive sleep apnea had nearly twice the risk of stroke or death over three years compared to those without the condition, even after accounting for other cardiovascular risk factors. Obstructive sleep apnea is a disease characterized by periodic, complete, or partial upper airway obstruction during sleep, causing intermittent pauses of breathing (apneas) or reductions in airflow (hypopneas) despite ongoing respiratory effort [43]. Another study showed that sleep-related breathing disorder appears to contribute as a risk factor for stroke [41]. Therefore, patients with sleep-related breathing disorders, which alter breathing patterns [39], are at a higher risk of stroke.

Stroke is more common among cigarette smokers. The impact of cigarette smoking on stroke incidence was assessed in the Framingham Heart Study cohort of 4255 stroke-free men and women who were aged 36 to 68 years [50]. During 26 years of follow-up, 459 strokes occurred, where the risk of stroke increased as the number of cigarettes smoked increased. Moreover, one study observed that smokers have a 50% higher risk of experiencing a stroke compared to non-smokers [44]. Another study showed that there were more decrements in respiratory capacity and control in the smokers group compared to the non-smokers [31]. A study also found that cigarette smokers exhibited 1.9 times more respiratory symptoms than non-smokers [38]. These findings highlight how the respiratory system behaves differently in smokers than in non-smokers, and thus may be used to assess the risk of stroke.

Breathing disorders are closely related to stroke-risk. A review article by Estol reviewed 32 published articles between 2002 and 2017 to assess the relation between air pollution and stroke [35]. The review observed that certain gaseous pollutants may induce thrombosis and alter the respiratory system of the patient, as well as highlighting that air pollution is a major risk factor for stroke [33, 35]. More studies were performed that show the direct link between stroke-risk and breathing disorders such as chronic obstructive pulmonary disease [30, 36, 40, 46], lung function complications

[42, 45, 47, 48], Coronavirus Disease 2019 (COVID-19) [34], and atrial fibrillation [49]. These respiratory complications have all been proven to be risk factors of stroke [37, 42]. This paves the way for using respiratory features as predictors of stroke-risk.

It remains unknown which signal or feature contributes the most in stroke-risk prediction, as AI models do not provide an in-depth analysis of how the model reached its output decision. Therefore, recent studies applied eXplainable AI (XAI) to provide interpretable results so humans can produce more explainable models [51], bridging the gap between an AI model's decision-making process and human comprehension [52]. This is especially important in the healthcare field where accurate real-time diagnosis and user trust are both important [53]. Some of the prominent XAI techniques are SHAP [54], LIME [55], and Anchors [56]. XAI was only applied to stroke-risk prediction models trained on individual signals [20, 21], failing to determine which signal plays the most critical role when several signals are used simultaneously.

SHAP, LIME, and Anchors are all post-hoc explainers, which means that they are used after the model is trained, and thus do not faithfully represent the true internal mechanisms of the model. However, they offer understandable insights into the model's decisions for human users. Additionally, they are simple to implement and do not compromise model accuracy or performance, which is crucial in this field. In contrast, intrinsically interpretable models generate more faithful representations of a model's internal mechanisms but are constrained by practical restrictions such as linearity, monotonicity, additivity, and sparsity. Current intrinsically interpretable models are typically simple and thus lack the flexibility to capture complex relationships that naturally occur in real-world applications [53, 57]. Post-hoc explainers also have the advantage of being model-agnostic, allowing them to be used even when the model becomes more complex, for example, with the availability of additional data.

The variability in outputs across different XAI models have raised the question of which model's interpretations are the most reliable, demonstrating the need for a standardized XAI evaluation metric that is objective and can be experimentally validated [58]. Elgezawy et al. proposed an XAI evaluation metric for classification: the Model Explanation Metric Consensus (MEMC) [59]. The XAI technique with the highest MEMC value is the most reliable. While this metric has not been applied widely, it acts as a fair evaluation metric that can be used.

In this study, we aim to address the research gaps in this field by offering the following contributions:

1. We integrated respiratory signals for the first time into a multi-modal stroke-risk prediction model alongside ECG, EMG, and blood pressure (BP) and discovered their importance in stroke-risk prediction, guiding future research into exploring the role of respiratory signals on stroke-risk and other cardiovascular diseases.
2. We applied three state-of-the-art XAI techniques namely LIME, SHAP, and Anchors to interpret the model's decisions and identify the most influential features in stroke-risk prediction and found that CO₂-derived features are key factors in stroke-risk prediction, guiding clinicians and medical experts to study the physiology of these signals further, and study the physiological aspect of these features.

3. We evaluated the reliability of these XAI techniques using the MEMC to determine which explainer offers the most trustworthy results, making this the first study to evaluate the use of XAI in this field.

The paper is structured as follows: Section 2 describes the data that was used in this study and the steps taken to conduct the experiment to allow for reproducibility. The results of the experiment are analysed in Section 3 and discusses them in Section 4, alongside the limitations and future work. Finally, Section 5 contains the conclusion which outlines the clinical significance of this work.

2 Methods

2.1 Dataset

The dataset that was used to conduct this study is available on Physionet [60] and was collected to investigate the effects of ischemic stroke on cerebral vasoregulation [61]. The dataset contains biological signals and general information of 60 stroke subjects and 60 non-stroke subjects. The control group consists of subjects with no clinical history of stroke and no focal deficit on a neurological exam. All stroke patients in this dataset experienced a stroke at least six months prior to their inclusion.

As part of the data collection process, subjects were asked to sit down for 5 minutes and then stand for another 5 minutes, during which signals were collected from their bodies. A Capnomac Ultima was connected to the subjects' mouths using an intubation tube and an airway adapter to measure end-tidal carbon dioxide (CO_2) and the respiration flow rate. Additionally, a Finapres device was used to measure BP. These three signals were recorded at a sampling rate of 500 Hz. ECG1 and ECG2 were recorded from the left and right clavicles of the subject, respectively. EMG1 and EMG2 were recorded from the left and right gastrocnemius muscles, respectively. The ECG and EMG were collected at a sampling rate of 1000 Hz using the ME6000 device. The seven signals are shown in Table 1 alongside the body location of each signal.

Table 1 Signals and the body locations from where they were collected from

Number	Signal	Location
1	ECG1	V5/V6-L Clavicle
2	ECG2	V1/V2-L Clavicle
3	EMG1	Gastrocnemius right
4	EMG2	Gastrocnemius left
5	BP	Finger
6	End-tidal carbon dioxide	Mouth
7	Respiration flow rate	Mouth

2.2 Overview

The methodology followed is depicted in Fig. 1. The data was first split into training and testing using an 80/20 ratio and stratification before any preprocessing to prevent data leakage, which would otherwise lead to overestimated accuracy. Stratified K -fold cross-validation is then performed on 80% of the data, which splits the data into K equal groups. One group is used for validation and the remaining $K - 1$ groups are used for training. This is repeated K times, with each group used for validation exactly once. Stratification ensures that the proportion of the two classes (stroke and non-stroke) are the same in both the validation and training sets, helping prevent class imbalance [62]. In this study, K was set to 5. The signals were windowed and features were extracted from each window. The features were then fed into the proposed model for training, and the model is evaluated on the unseen test set. Finally, the model's predictions on the test set were fed into the XAI models, which were then evaluated using the MEMC.

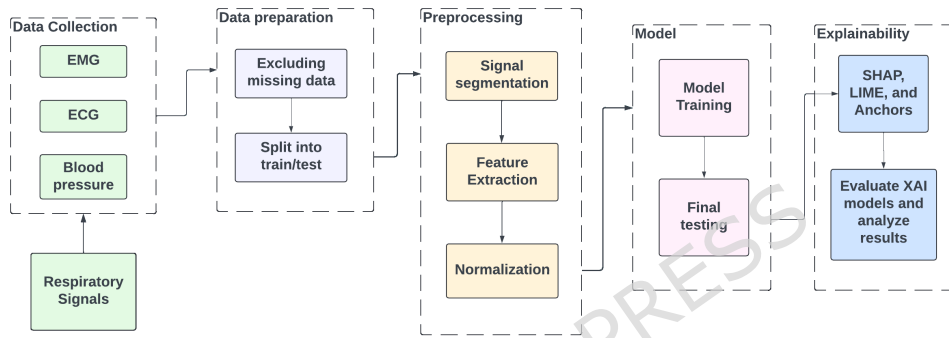


Fig. 1 Overview of methodology

2.3 Preprocessing

2.3.1 Signal Windowing

Stroke-risk prediction models used various window sizes as seen in Section 1. To find the optimal window size and overlap percentage, we performed a parameter search of window sizes and overlap values. One study using respiratory signals tested window sizes of 5 to 15 seconds, with overlap values of 10%, 25%, 50%, 75%, and 90%, to classify different breathing patterns and respiration disorders [63]. We included smaller window sizes of 1 to 5 seconds as ECG and EMG were found to be more effective when windowed at these values [64–67]. Therefore, we tested window sizes of 1 to 15 seconds, with a step size of 1 second, for each overlap value mentioned. The resulting dataset consists of samples, and each sample in the dataset is made up of seven windows.

2.3.2 Feature Extraction

For each sample, statistical and frequency-domain features were extracted from each of the seven signals. Features are values that can give insights into the signal's behavior and pattern. Additionally, using features for training a model uses less computational resources than using raw signals. The list of features extracted is shown, where items 1–11 are the statistical features and items 12–15 are frequency-domain features. Statistical features are commonly used for biological signal analysis [68, 69] and are able to extract valuable information [70]. Moreover, biological signals are non-linear, non-stationary, and non-Gaussian. Higher order statistics such as skewness, kurtosis, 3rd, 4th, and 5th moments are also extracted which are able to detect and capture the non-linear properties of these signals, while being less sensitive to their dynamic nature [71]. While these features do not directly relate to physiological measures, they capture the irregularities in the signal that may reflect underlying physiological changes [72, 73]. The zero-crossing feature is extracted as it is a measure of frequency information in the time-domain, and is used to assess the rapidity of signal fluctuations which can also reflect physiological irregularities [74]. Frequency-domain features capture the signal's periodic properties and spectral distribution, which may not be captured by statistical features, and also reflect activity in the sympathetic and parasympathetic nervous systems [75], which is important for identifying stroke-risk [76].

1. Sample mean, $\bar{y}$

$$\bar{y} = \frac{1}{N} \sum_{n=0}^{N-1} y[n] \quad (1)$$

where N is the total number of data points in a window y , and $y[n]$ is the n^{th} data point in y .

2. Sample standard deviation [77], s

$$s = \sqrt{\frac{1}{N-1} \sum_{n=0}^{N-1} (y[n] - \bar{y})^2} \quad (2)$$

3. Sample variance, s^2

4. Median [68], which is the mid-point value of y ,

$$\text{median}(y) = \begin{cases} y[\frac{N+1}{2}], & \text{if } N \text{ is odd} \\ \frac{y[\frac{N}{2}] + y[\frac{N}{2} + 1]}{2}, & \text{if } N \text{ is even} \end{cases} \quad (3)$$

5. Sample skewness [78], G_1

$$G_1 = \frac{\sqrt{N(N-1)}}{N-2} \frac{\frac{1}{N} \sum_{n=0}^{N-1} (y[n] - \bar{y})^3}{(\frac{1}{N-1} \sum_{n=0}^{N-1} (y[n] - \bar{y})^2)^{\frac{3}{2}}} \quad (4)$$

6. Sample kurtosis [78], G_2

$$G_2 = \frac{N(N+1)(N-1) \sum_{n=0}^{N-1} (y[n] - \bar{y})^4}{(N-2)(N-3)(\sum_{n=0}^{N-1} (y[n] - \bar{y})^2)^2} - \frac{3(N-1)^2}{(N-2)(N-3)} \quad (5)$$

7. q^{th} Moment [68], $M^{(q)}$,

$$M^{(q)} = \frac{1}{N} \sum_{n=0}^{N-1} y[n]^q, q \in \{3, 4, 5, 6\} \quad (6)$$

8. Minimum [68], which is the smallest value in y .

9. Maximum [68], which is the largest value in y .

10. Zero Crossings [68], which is the amount of times the sign changed in y , i.e. from negative to positive or vice versa

$$ZeroCrossings = \sum_{n=0}^{N-2} \mathbf{1}(y[n] \times y[n+1] < 0) \quad (7)$$

11. Mean energy [68], E_{mean}

$$E_{mean} = \frac{1}{N} \sum_{n=0}^{N-1} y[n]^2 \quad (8)$$

12. Fast Fourier Transform (FFT) energy [79], E_{FFT} , which measures the total energy of the signal in the frequency-domain and is obtained by summing the squared magnitudes of all Fourier coefficients

$$E_{FFT} = \sum_{m=0}^{N-1} |Y[m]|^2 \quad (9)$$

where $Y[m]$ is the m^{th} Fourier coefficient [79], seen in Eq. (10)

$$Y[m] = \sum_{n=0}^{N-1} y[n] \times e^{-i \frac{2\pi}{N} mn}, \quad m = 0, 1, \dots, N-1, \quad i = \sqrt{-1} \quad (10)$$

13. Spectral entropy [80], H

$$H = - \sum_{m=0}^M PSD_{\text{norm}}[m] \log_2(PSD_{\text{norm}}[m]), \quad M = \frac{N}{2} + 1 \quad (11)$$

where $PSD_{\text{norm}}[m]$ is the m^{th} normalized power spectral density (PSD) [80]

$$PSD_{\text{norm}}[m] = \frac{PSD[m]}{\sum_{m=0}^M PSD[m]} \quad (12)$$

415 PSD_m is the m^{th} single-sided power spectral density calculated using Eq. (13)

416

$$417 \quad PSD[m] = \begin{cases} \frac{2}{N \times f_s} \times |Y[m]|^2, & m = 1, 2, \dots, M-1 \\ \frac{1}{N \times f_s} \times |Y[m]|^2, & m = 0, M \end{cases} \quad (13)$$

419

420 and f_s is the sampling frequency.

421 14. Low frequency power [81], P_{low} , which is the sum of all $PSD[m]$

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$$423 \quad P_{\text{low}} = \sum_m PSD[m], \quad f[m] \in [0.04, 0.15) \quad (14)$$

424

425 where m corresponds to frequencies $f[m]$ within the range $[0.04, 0.15)$ and are

426 found using Eq. (15)

$$427 \quad f[m] = \frac{m}{N} \quad (15)$$

428

429 15. High frequency power [81], P_{high} , which is the sum of all $PSD[m]$ corresponding

430 to frequencies $f[m]$ within the range $[0.15, 0.4)$

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$$432 \quad P_{\text{high}} = \sum_m PSD[m], \quad f[m] \in [0.15, 0.4) \quad (16)$$

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434 For each sample in the dataset, the features were extracted from each of the seven

435 signals and were fused together to form a feature set x .

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2.3.3 Normalization

We performed mean-based normalization on the extracted features, which causes each feature to have a mean of 0 and a standard deviation of 1 between all samples. The new normalized data point $x[n]$ was computed using Eq. (17),

$$444 \quad x[n] = \frac{x_{\text{initial}}[n] - \bar{x}_p}{s_p} \quad (17)$$

447 where $x_{\text{initial}}[n]$ is the initial feature value, and $\bar{x}_p$ and s_p are the mean and standard deviation values for the p^{th} feature, respectively, between all samples.

2.4 Model

451 We used a single-layer perceptron for this study, as shown in Fig. 2. The perceptron computes a weighted sum of the input features and adds a bias term. This result is then passed through a sigmoid activation function, which transforms it into a probability between 0 and 1. A threshold of 0.5 is applied: if the output probability exceeds 0.5, the model predicts a stroke. Otherwise, it predicts non-stroke.

2.5 Model Evaluation

459 We compared our model with recent models used in this field [12, 15, 16, 19, 26] on the same dataset with the same random seed. The models that were tested against

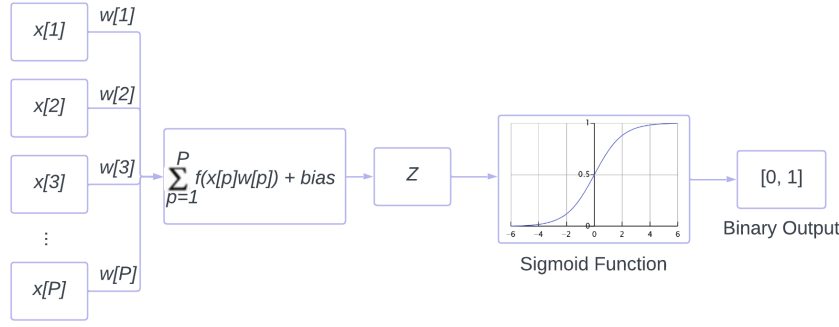


Fig. 2 A single perceptron layer where $x[p]$, $w[p]$, are the inputs and weights for the p^{th} feature, respectively. Z is the output probability of the model which is converted to a binary output

are: CNN, LSTM, BiLSTM, CNN-LSTM [16], CNN-BiLSTM [12], E-ESN [19], LSTM [15], XGBoost, Random Forest, Logistic Regression, KNN, Decision Tree, SVC, and the voting classifier [26]. Each model was evaluated on the unseen test set, and the performance metrics were computed based on the true positives (TP), true negatives (TN), false positives (FP), false negatives (FN), as in Eqs. (18)–(21).

$$Accuracy(\%) = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (18)$$

$$Precision(\%) = \frac{TP}{TP + FP} \times 100 \quad (19)$$

$$Recall(\%) = \frac{TP}{TP + FN} \times 100 \quad (20)$$

$$Specificity(\%) = \frac{TN}{TN + FP} \times 100 \quad (21)$$

Furthermore, to analyze the effect of incorporating respiratory signals into stroke-risk prediction models, each model was evaluated twice: once with respiratory signals and once without. We also conducted a statistical t -test on our proposed model when trained with and without respiratory signals using the validation results of the K -fold cross validation with $K = 5$. This helps determine whether incorporating respiratory signals yields significant improvements. We calculate the t -values using Eq. (22),

$$t = \frac{\bar{d}}{s_d / \sqrt{R}} \quad (22)$$

where R is the number of pairs, $\bar{d}$ is the mean of the differences between paired observations and s_d is the standard deviation of those differences which are calculated using Eqs. (23)–(25),

$$\bar{d} = \frac{1}{R} \sum_{r=1}^R d[r] \quad (23)$$

$$s_d = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (d[r] - \bar{d})^2} \quad (24)$$

$$d[r] = acc_{\text{with}}[r] - acc_{\text{without}}[r], \quad r = 1, \dots, R, \quad R = K = 5 \quad (25)$$

where $d[r]$ is the difference between each pair, $acc_{\text{with}}[r]$ and $acc_{\text{without}}[r]$ are the model's accuracy when trained with and without respiratory signals, respectively, at the r^{th} fold. The calculated t -value was used to find the p -value from the t -distribution table. We used a significance level (α) of 0.05. If the p -value is smaller than α , the results are statistically significant.

2.6 XAI

2.6.1 SHAP

SHAP is an XAI method based on game theory [82] that provides global and local explanations. Local explanations study the reasoning behind the model's prediction on a specific instance while global explanations explain the behaviour of the model across the entire dataset. SHAP works by removing a feature and evaluating how its removal affects the model's performance using SHAP values, which quantify the feature's contribution on the model's predictions [54].

For a trained model f , the output prediction of the model $f(x)$ for the input x in the test set is explained using Eq. (26),

$$f(x) = E[f(X)] + \sum_p \phi[p] \quad (26)$$

where $E[f(X)]$ is the average output of the model on the training set, X , with N_{train} samples and is calculated in Eq. (27)

$$E[f(X)] = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} f(X[i]) \quad (27)$$

and $f(X[i])$ is the prediction on the i^{th} sample in X . $\sum_p \phi[p]$ is the sum of the contributions from all features for that instance, where $\phi[p]$ is the SHAP value for the p^{th} feature and is computed as in Eq. (28) [54]

$$\phi[p] = \sum_{S \subseteq F \setminus \{p\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{p\}}(x_{S \cup \{p\}}) - f_S(x_S)] \quad (28)$$

where $f_{S \cup \{p\}}$ refers to the trained model with the p^{th} feature present, and f_S is the trained model without that feature. The contribution of the p^{th} feature is estimated by taking the difference in predictions: $f_{S \cup \{p\}}(x_{S \cup \{p\}}) - f_S(x_S)$, where x_S represents the values of the input features in the set S . Since the effect of withholding a feature depends on other features in the model, the difference is calculated across all possible subsets $S \subseteq F \setminus \{p\}$, where F is the set of all features.

A positive SHAP value indicates that the feature increases the likelihood of a model predicting the positive class (stroke) while a negative value decreases it. The magnitude of the SHAP value reflects the strength of the feature's contribution: the higher the absolute value, the more impactful that feature was in the model's prediction. SHAP global explanations are achieved by averaging the SHAP values for all instances.

2.6.2 LIME

LIME is a local explainer that interprets individual predictions by approximating the behaviour of a complex model f around a specific instance x [55]. It works by creating a local model g that approximates the predictions of the model f in the neighbourhood of x . To do this, it generates perturbed samples x' around x , and obtains their predictions from f , and then represents x' in an interpretable binary form. The surrogate model g is then trained on these binary samples, with each one weighted by its proximity to x using a kernel π_x by solving the optimization problem in Eq. (29)

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (29)$$

where $\mathcal{L}(f, g, \pi_x)$ is the loss function that measures how well the simple model g approximates the original model f near x , π_x is a proximity kernel that assigns higher weights to samples that are closer to the original input x in the input space, $\Omega(g)$ is a regularization term that penalizes the complexity of the model g to promote simplicity, and G is a set of interpretable models. The explanation model $\xi(x)$ for x is obtained using the $\operatorname{argmin}_{g \in G}$ argument, which finds the model $g \in G$ that minimizes the sum of the local loss $\mathcal{L}(f, g, \pi_x)$ and the complexity penalty $\Omega(g)$.

The optimization in Eq. (29) therefore selects the best local surrogate model that balances accurately approximating the original model f near x with being simple and interpretable. The weights assigned to each feature by this model represent how much that feature contributes to the prediction made by the original model f . A positive weight indicates that the feature increases the likelihood of the model predicting a stroke, while a negative weight decreases it.

2.6.3 Anchors

Anchors [56] explains the behaviour of a black-box model $f(x)$ for a specific input x by identifying high-precision if-then rules. An anchor A is a set of rules that, when satisfied by the input x , are enough to make the model keep giving the same prediction, even when other features are changed through perturbations. Perturbations are sampled from the conditional distribution $D(z|A)$, which represents the conditional distribution over perturbed samples z , where only the features not included in the anchor A are varied. An anchor is accepted if the precision condition in Eq. (30) is satisfied,

$$\mathbb{E}_{D(z|A)}[1_{f(z)=f(x)}] \geq \tau, A(x) = 1 \quad (30)$$

where $A(x) = 1$ means that the anchor applies to input x , i.e., all the conditions in the rule are satisfied by x . The indicator function $1_{f(z)=f(x)}$ counts how often the

prediction on the perturbed sample z matches the original prediction on x . The left-hand side of the inequality computes the precision of the anchor which is the proportion of perturbed samples for which the model's prediction remains unchanged. An anchor is accepted if this precision is greater than or equal to a user-defined threshold τ , which we set to 0.95 as in [56].

2.6.4 XAI Evaluation

The MEMC [59] was computed to determine which XAI model is the most trusted. Firstly, the feature that was deemed most important by the XAI model is removed from the original dataset and the new evaluation metrics are computed, as in Eqs. (18)–(21). Secondly, the two most important features are removed and the new performance metrics computed. Thirdly, the three most important features are removed and the new performance metrics computed. The MEMC is then computed by summing the difference of the initial metrics and the new metrics, as shown in Eq. (31),

$$MEMC = \frac{1}{V} \sum_{v=1}^V [(accuracy_o - accuracy[v]) + (precision_o - precision[v]) + (recall_o - recall[v]) + (specificity_o - specificity[v])], V = 3 \quad (31)$$

where $accuracy[v]$, $precision[v]$, $recall[v]$, and $specificity[v]$ are the new computed metrics after removing v features from the original dataset, and V refers to the total amount of top features to be removed. $accuracy_o$, $precision_o$, $recall_o$, and $specificity_o$ are the metrics computed with all features present. The XAI model that results in the highest MEMC value is the model to be trusted the most.

3 Results

3.1 Data Analysis

Subjects with missing signals or incomplete tests were excluded from this study, leaving 64 subjects, 37 of which suffered a stroke. From the 10 minutes of data collected, the signals were trimmed to 9.5 minutes, as some subjects had missing data in the last 30 seconds. ECG and EMG were down-sampled to 500 Hz to match the other signals so that the extracted features can be fairly fused together. Demographic information regarding the remaining subjects is seen in Table 2. To determine whether the two groups come from the same population, independent t -tests were conducted using the IBM SPSS Statistics software on Age, BMI, Height, and Mass, with $\alpha = 0.05$. The resulting p -values are in Table 2. Since all the p -values are greater than α , there is no significant difference between the two groups.

3.2 Optimal Window Size and Overlap Percentage

The performance of the proposed model when tested with different window sizes and overlap values is in Fig. 3. The worst performance was achieved when a window size of 1 second was used, where the resulting accuracies are less than 66% for all overlap

Table 2 Participant demographics of both groups, and results of statistical *t*-tests. Continuous values are written as mean $\pm$ standard deviation

Feature	Stroke group	Control group	<i>p</i> -value
Sex (female, male)	19, 18	15, 12	-
Age (years)	65.16 $\pm$ 8.60	66.89 $\pm$ 6.79	0.9368
BMI (kg/m ²)	27.72 $\pm$ 5.04	25.52 $\pm$ 4.53	0.3908
Height (m)	1.66 $\pm$ 0.08	1.69 $\pm$ 0.10	0.2412
Mass (kg)	76.87 $\pm$ 15.24	72.88 $\pm$ 13.80	0.2860

values. Increasing the window size increased accuracy generally, and this was seen in all overlap values. However, the highest accuracy of 79.87% was achieved when using a window size of 14 seconds and an overlap of 75%. The window size of 14 seconds was also found to be the most effective in classifying breathing disorders [63], yet was achieved when overlap was set to 90%. Therefore, a window size of 14 seconds and an overlap of 75% were used for this study.

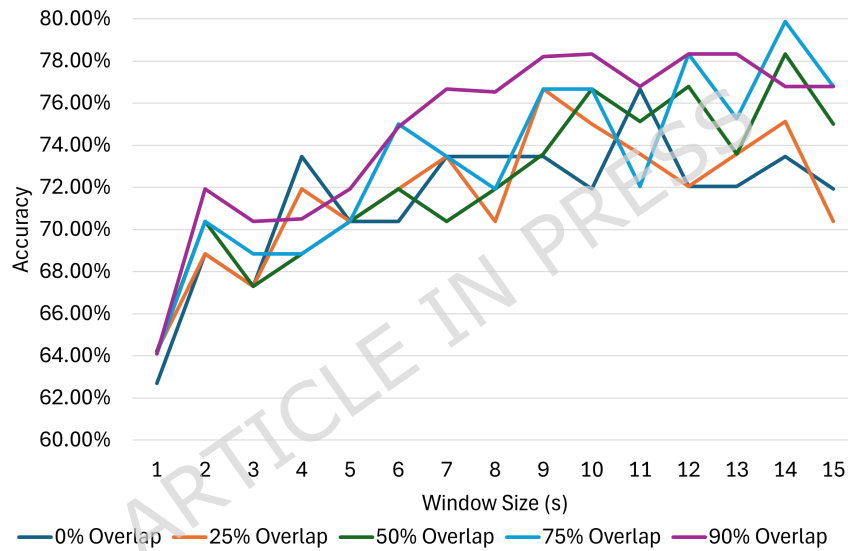


Fig. 3 Accuracy of proposed model for different window sizes and overlap values

3.3 Model Evaluation

Using the optimal window size and overlap values found above, we performed a grid search to find the optimal hyperparameter values for the model. It was found that

increasing the number of layers degraded our model's performance. Therefore, a single-layer perceptron was used. The highest accuracy of 84.91% was obtained when Adam optimizer was used and the model was trained for 700 epochs, with a learning rate of 0.03 and a batch size of 128. The model was tested with and without respiratory signals to assess their significance. The mean accuracy of the proposed model was $84.91\% \pm 5.48\%$ with respiratory signals and $59.36\% \pm 12.15\%$ without them, as shown in Fig. 4. We conducted a t -test at $\alpha = 0.05$ and found the p -value to be 0.0061, which is less than α . Therefore, there is significant difference between the two cases.

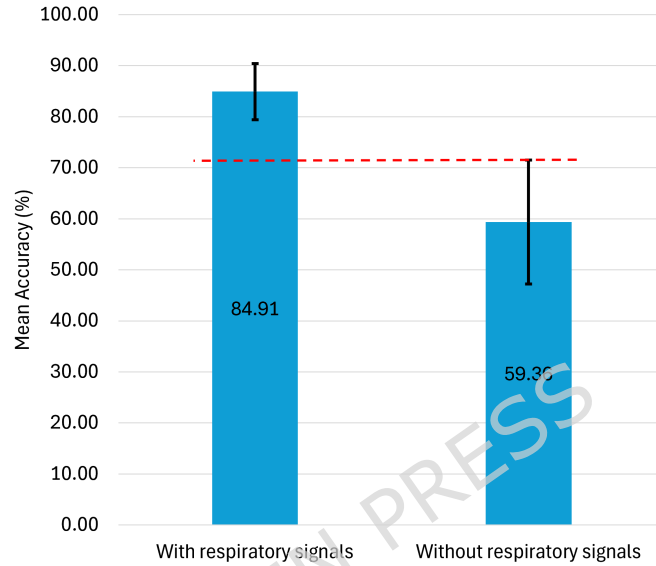


Fig. 4 Mean accuracy of model with and without respiratory signals with standard deviation error bars. The error bars do not overlap which supports the significant difference between both cases

To validate the model's performance, we compared our proposed model with recent ML and DL models. We reproduced each study's method as closely as possible while using the same dataset discussed in Section 2.1 [60]. For techniques relying on time-series data [12, 15, 16], we used the raw signals. For other studies using features [19, 26] and ML methods, we extracted the features in Section 2.3.2. The results of each model's performance on the unseen test set are in Fig. 5, where our proposed model achieved the highest accuracy of 88.29% with respiratory signals. Incorporating respiratory signals generally improved performance, highlighting their value in stroke-risk prediction. Only a few models, namely BiLSTM, E-ESN, and XGBoost, performed better without respiratory signals, though the differences were small and the performance remained poor. On the other hand, adding respiratory signals improved accuracy by over 10% for the Voting Classifier [26], Logistic Regression, KNN, and our model.

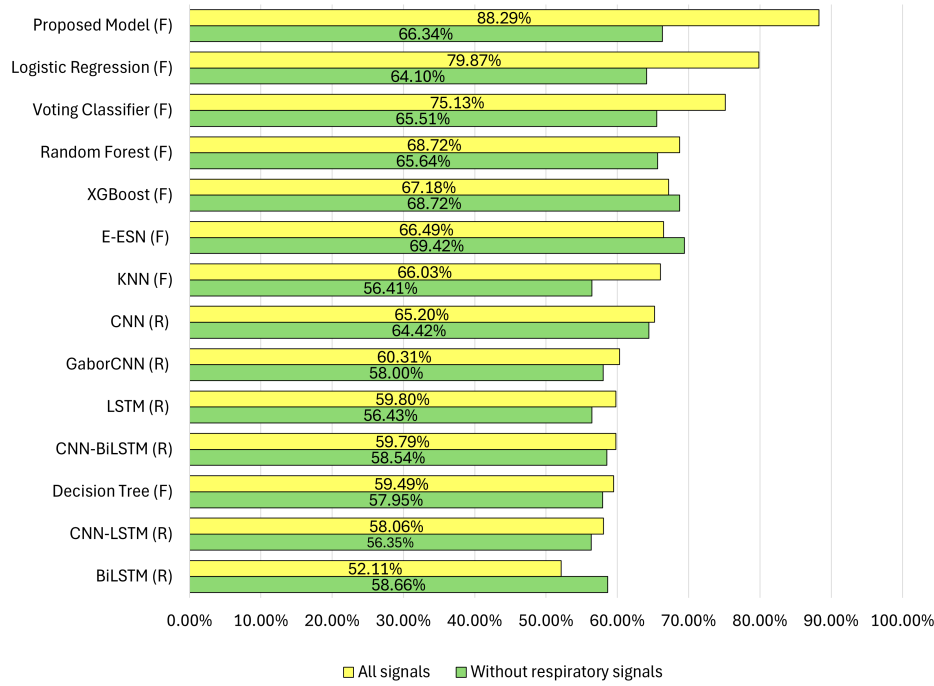


Fig. 5 Performance of proposed model against other ML and DL models. The letter (F) or (R) after each model describes whether features or raw signals were used, respectively

3.4 XAI

3.4.1 SHAP

The SHAP global summary plot showing the 12 most important features, 8 of which are derived from respiratory signals (CO_2 and flow rate), is in Fig. 6. According to SHAP, the most important feature is the CO_2 mean which changed the model's predicted probability by 0.15 on average. The CO_2 variance and BP mean (abp_mean) were also key predictors and changed the model's output by approximately 0.14, on average.

The SHAP summary plot in Fig. 7 shows: (1) the direction in which these features influenced the model's prediction (stroke or non-stroke), and (2) which values of these features (low or high) caused the model to move in that direction. A high CO_2 mean increased the probability of predicting stroke by up to 0.75, and a low value decreased this probability by up to 0.75. Additionally, a high abp_mean and a low CO_2 variance were influential factors in predicting stroke.

Local SHAP plots for a stroke and a non-stroke subject are seen in Fig. 8. The plots start at $E[f(X)]$ which represents the model's baseline output for all subjects and is calculated using Eq. (27). Each feature's SHAP value is then added to $E[f(X)]$ to achieve the final prediction for that subject, as in Eq. (26). Since $E[f(X)]$ is larger

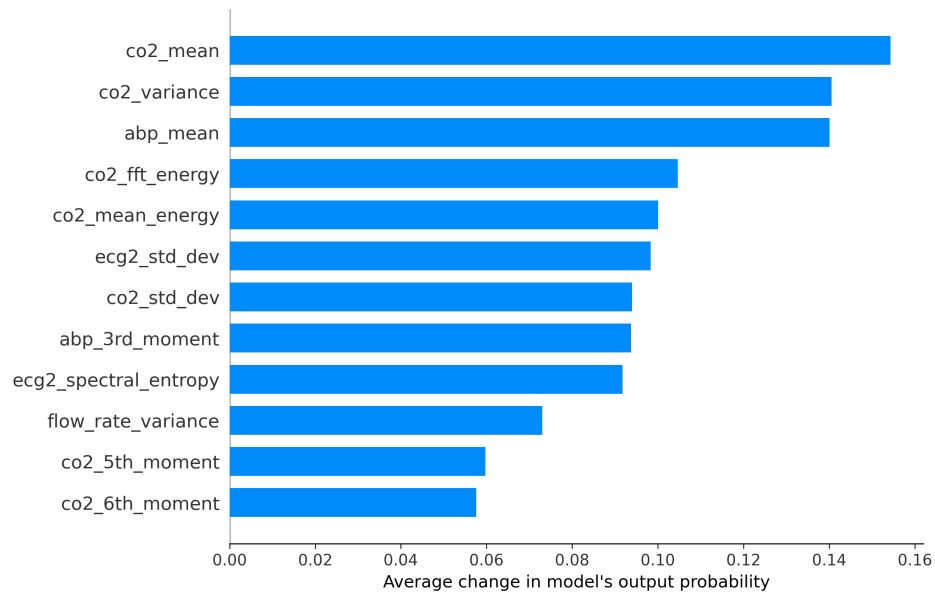


Fig. 6 SHAP summary plot showing the 12 most important features. The horizontal axis shows the average absolute SHAP value. std_dev: standard deviation

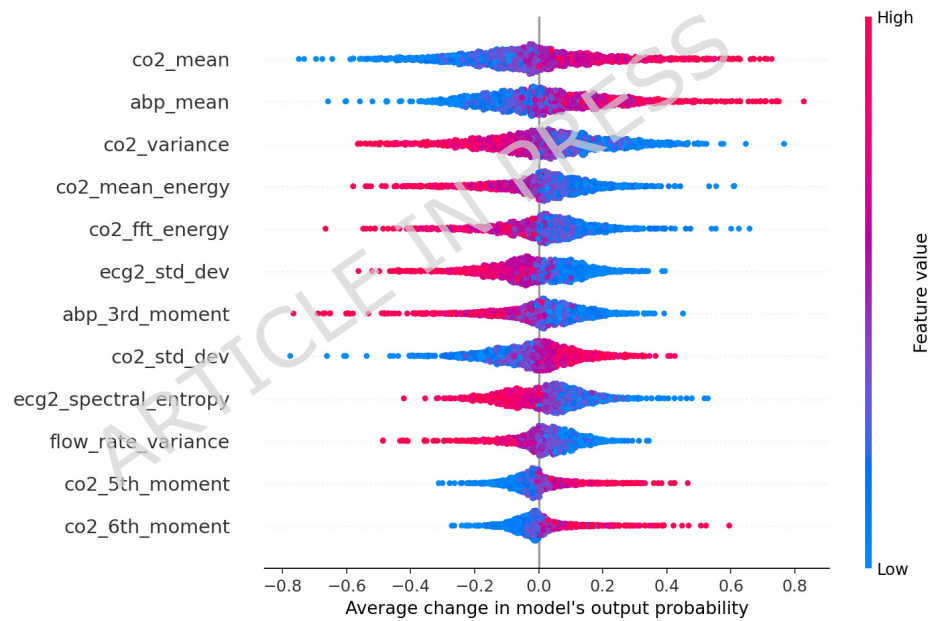


Fig. 7 SHAP summary plot with feature effects. Points right of the line increase the model's likelihood in predicting a stroke, whereas points left of the line decrease it. The points' colours represent the magnitude of the feature value, where pink is a high feature value and blue is a low feature value

than 0.5, the model's baseline prediction is a stroke, this bias may have occurred as there are more stroke subjects in the training dataset than control.

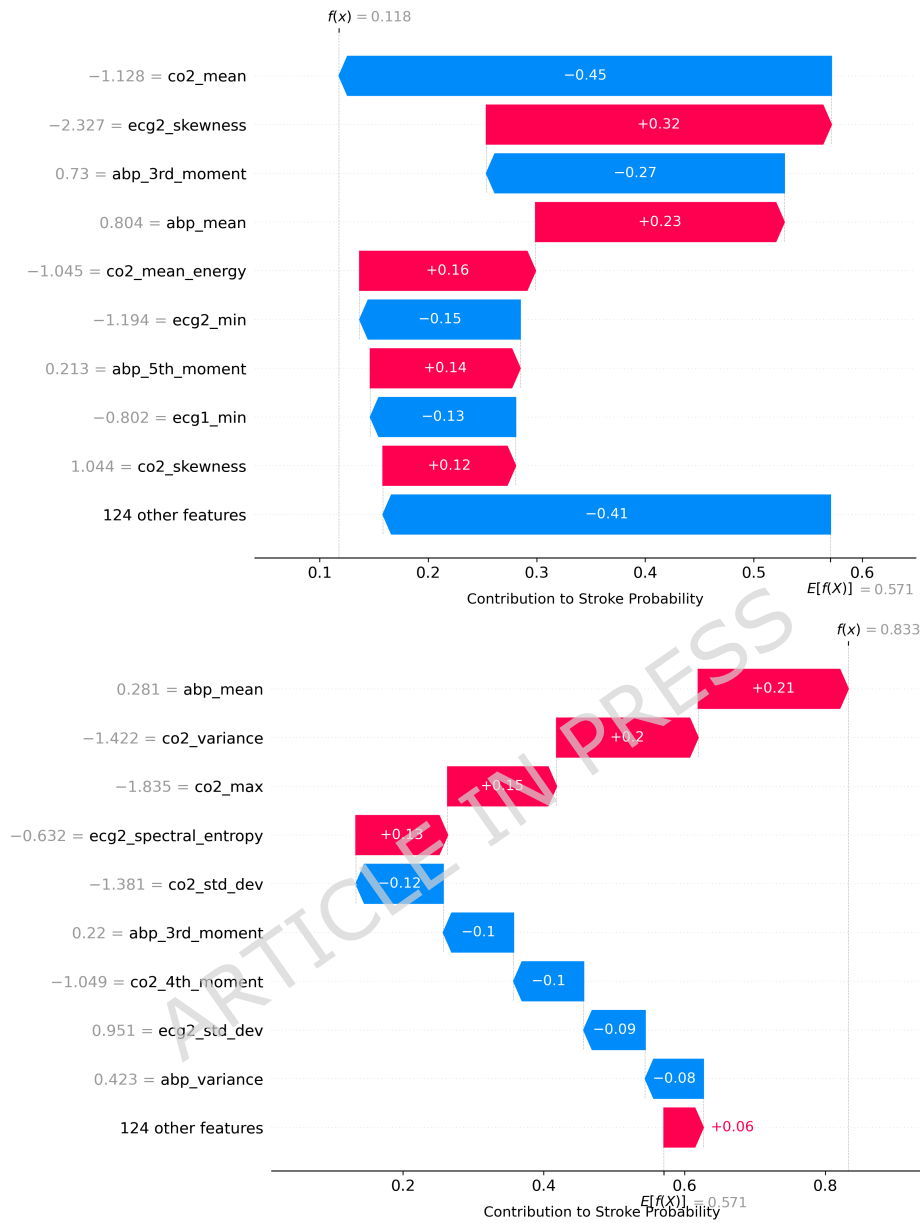


Fig. 8 Top: Waterfall plot for a non-stroke subject with $f(x)$ as 0.118. Bottom: Waterfall plot for a stroke subject with $f(x)$ as 0.833. The waterfall plots show how SHAP values affected the model's output probability $f(x)$, depending on feature values. The horizontal axis represents the probability of predicting a stroke and the vertical axis shows the features and their values

For the non-stroke subject in Fig. 8, the least important features referred to as “124 other features” when combined decreased the predicted probability of stroke by 0.41. The largest contribution towards predicting non-stroke was caused by the CO₂ mean value of -1.128, which reduced the model’s probability of predicting a stroke by 0.45. This is consistent with the global SHAP findings in Fig. 7. The second most influential feature was the ECG2 skewness with a value of -2.327 which increased the prediction of a stroke by 0.32. However, after adding all SHAP values to the base value, $E[f(X)]$, the final predicted probability is 0.118, resulting in a non-stroke prediction. For the stroke subject, the low values of 0.281 and -1.422 for the BP mean and CO₂ variance, respectively, were the most influential in causing the model to predict the stroke class, with SHAP values of 0.21 and 0.2, respectively. Finally, the SHAP values combined for this subject resulted in a final probability of 0.833, leading to a stroke prediction.

3.4.2 LIME

The LIME global plot was created by aggregating the absolute LIME weights of each feature across all test instances [55]. The 12 most important features according to LIME, are in Fig. 9, where the three most important features are the amount of zero crossings from CO₂ (co2_zero_crossings), the mean flow rate, and the CO₂ variance.

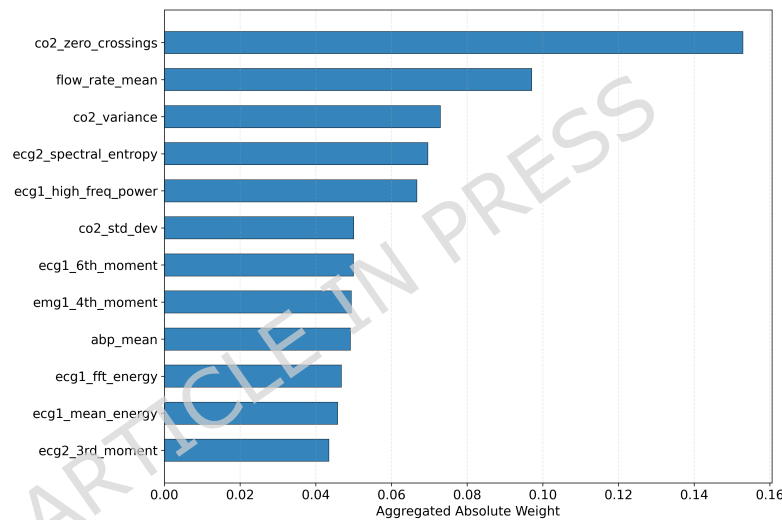


Fig. 9 The 12 most important features according to LIME. The horizontal axis shows the aggregated absolute weight for each feature

The LIME explanations for a stroke and a non-stroke subject are in Fig. 10. For the non-stroke subject, the amount of zero crossings by BP (abp_zero_crossings) being less than or equal to -0.03 contributed the most to predicting a non-stroke with an assigned weight of 0.198. The three next most contributing features towards correctly predicting non-stroke were the ECG1 mean energy, CO₂ zero crossings, and the ECG1 FFT energy. For the stroke subject, the ECG1 high frequency power being greater than

0.08 had the largest positive weight, i.e. contributed the most towards the prediction of stroke. The other influential features in correctly predicting stroke for this subject were the means of the ECG2, ECG1, and EMG1.

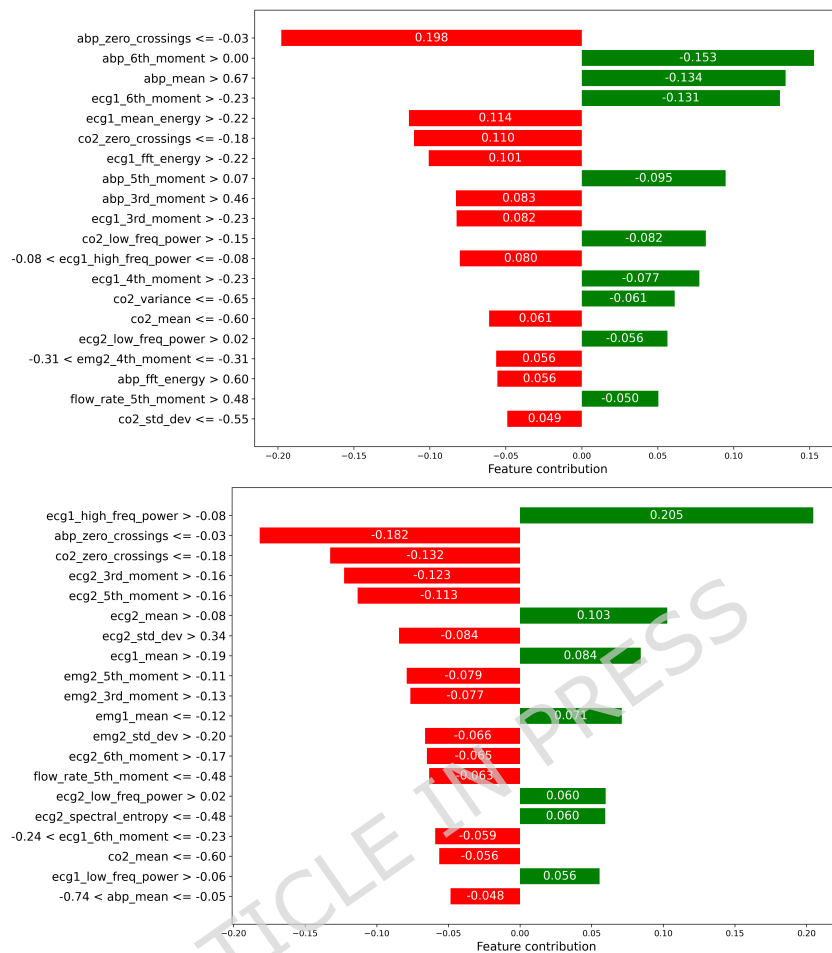


Fig. 10 LIME explanation of a correctly predicted non-stroke subject (top) and a stroke subject (bottom). Each bar represents the weight assigned by the surrogate model g to that specific feature, and the bar's length corresponds to the strength of that feature. Green bars indicate an increase in the likelihood of a stroke prediction, whereas red bars represent the opposite

3.4.3 Anchors

To create a global plot from Anchors, we counted how often each feature appears in a rule for all predictions (Fig. 11). This was done following SHAP's and LIME's method of computing global importance, which is to aggregate the feature importance of each feature [54, 55]. The CO₂ mean was the most influential feature according

to Anchors, appearing 61 times as part of an Anchor. The second and third most important features were the EMG2 median and ECG2 mean, respectively.

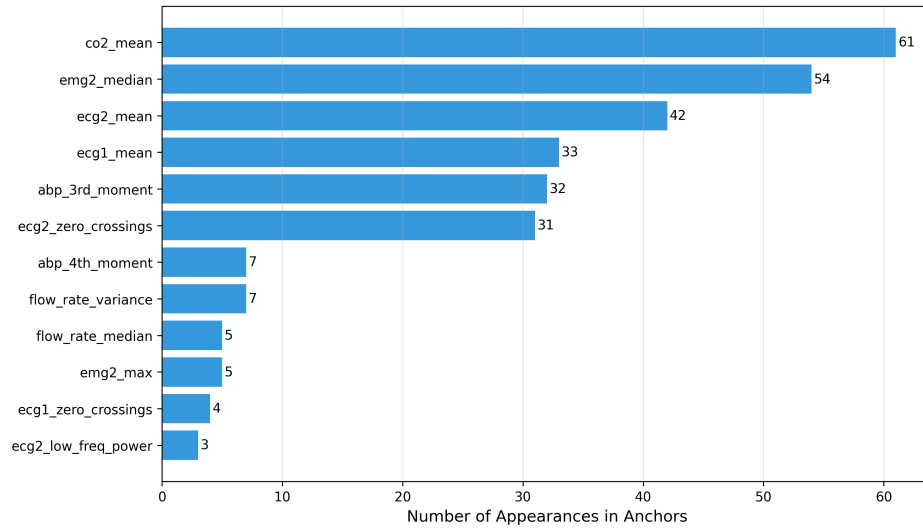


Fig. 11 Feature importance ranked by Anchors. The horizontal axis refers to how many times each feature appeared in an Anchor

The anchors for a non-stroke and a stroke subject are seen in Fig. 12. For the non-stroke prediction, the anchor identified three rules that when satisfied, were sufficient for the model to classify the subject as non-stroke; a flow rate skewness greater than 1.09, an ECG2 kurtosis greater than 0.42, and a flow rate variance greater than 1.28. The corresponding feature values for this subject were 1.44, 0.69, and 2.02, respectively. On the other hand, the set of rules for predicting stroke for the stroke subject was a low EMG2 median and BP 3rd moment.

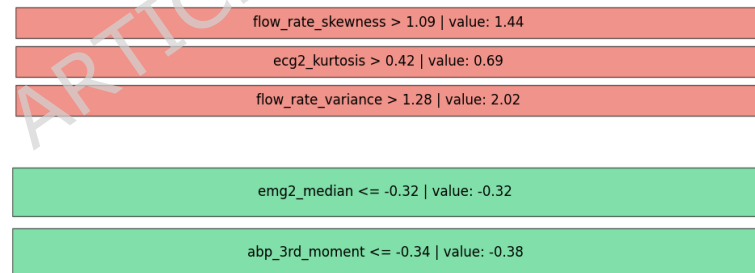


Fig. 12 Local Anchors explanations. Top: Anchors for the non-stroke subject represented by the red bars. Bottom: Anchors for the stroke subject represented by the green bars

3.4.4 XAI Evaluation

The three most influential features according to each XAI model are in Table 3. SHAP and Anchors agreed that the CO₂ mean was the most important feature. SHAP and LIME also agreed on CO₂ variance as one of the most important feature, although the order differs. Each XAI model identified at least one feature derived from respiratory signals as important. The top three most important features according to LIME are all derived from respiratory signals. Only Anchors identified ECG and EMG features as some of the most important.

Table 3 Top three important features according to each XAI model

Feature Ranking	SHAP	LIME	Anchors
1	CO ₂ mean	CO ₂ zero crossings	CO ₂ mean
2	CO ₂ variance	Flow rate mean	EMG2 median
3	BP mean	CO ₂ variance	ECG2 mean

As seen in Table 3, the three XAI models produced different results, thus creating doubts regarding which results to follow. To address this, we find the most reliable model by selecting the one with the highest MEMC value. To calculate the MEMC of each XAI model, the 3 most important features according to each model must be removed in order, and the new metrics must be computed each time a feature is removed, as in Eq. (31). In Tables 4–6, the metrics after removing the most important features are observed for SHAP, LIME, and Anchors, respectively. Removing the three most important features according to each XAI model noticeably degraded performance. Removing the CO₂ mean, which is the most important feature according to SHAP and Anchors, decreased the accuracy by 1.88% (Table 4 and Table 6). On the other hand, removing the CO₂ zero crossings which is the most important feature according to LIME, reduced the accuracy by only 0.63% (Table 5). Removing the two most important features according to the three XAI models decreased the accuracy by 2.71% for SHAP, 1.88% for LIME, and 2.13% for Anchors. Lastly, removing the three most important features decreased the accuracy by 3.19% for SHAP, 1.88% for LIME, and 2.03% for Anchors.

Table 4 Performance metrics after removing most important features according to SHAP

Features Removed	Accuracy	Precision	Recall	Specificity
No feature removed	88.29%	94.94%	85.53%	92.70%
Removing the most important feature	86.70%	93.61%	84.12%	90.82%
Removing the top two most important features	85.58%	94.68%	81.13%	92.70%
Removing to top three most important features	85.10%	92.13%	82.86%	88.86%

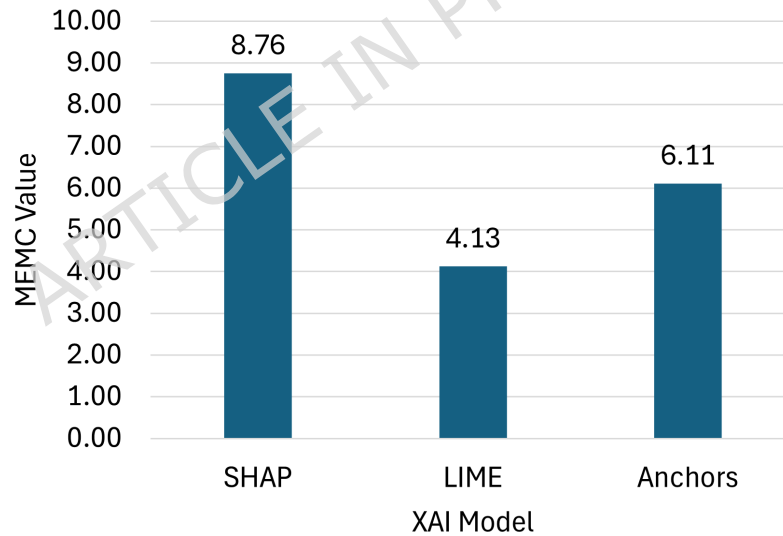
Table 5 Performance metrics after removing most important features according to LIME

Features Removed	Accuracy	Precision	Recall	Specificity
No feature removed	88.29%	94.94%	85.53%	92.70%
Removing the most important feature	87.66%	94.64%	84.75%	92.33%
Removing the top two most important features	86.41%	95.84%	81.45%	94.34%
Removing to top three most important features	86.41%	93.66%	83.57%	90.94%

Table 6 Performance metrics after removing most important features according to Anchors

Features Removed	Accuracy	Precision	Recall	Specificity
No feature removed	88.29%	94.94%	85.53%	92.70%
Removing the most important feature	86.70%	93.61%	84.12%	90.82%
Removing the top two most important features	86.16%	95.65%	81.21%	94.09%
Removing to top three most important features	86.26%	93.33%	83.65%	90.44%

The MEMC for each XAI model is seen in Fig. 13. The results show that SHAP is the most reliable XAI model as observed by the highest MEMC value. Therefore, the three most important features in our model are the same ones identified by SHAP, which are the CO₂ mean, CO₂ variance, and the BP mean.

**Fig. 13** MEMC for each XAI model

4 Discussion

The first step was to find the optimal window size along with the overlap percentage, which we found to be 14 seconds and 75% overlap. Physiological signals such as ECG, blood pressure, EMG, and respiration are quasi-periodic and inherently noisy. Very short windows are highly sensitive to noise, leading to unstable time- and frequency-domain feature estimates. When only a single cycle of a biological signal is captured, extracted features may be dominated by local irregularities rather than representative signal behavior. On the other hand, longer windows may introduce redundant information and reduce temporal sensitivity and dilute transient but informative patterns relevant to the classification task. With a window size of 14 seconds, at least 3 breathing cycles were captured, and the model could relate these cycles with the other signals. This also means that the model prioritised capturing the respiratory signals' cycles, also proving that they are important in this task. Even when enough cycles of ECG, EMG, and BP were captured, the model still performed the best when a few breathing cycles were captured. A recent study has also demonstrated that 14-second windows are effective for respiratory disorder diagnosis [63]. Capturing multiple cycles reduces the influence of noisy individual breaths and yields more stable physiological representations. Additionally, the optimal window size depends on the classification task. Longer windows have been shown to be preferable for tasks such as apnea detection [83] or certain ECG analyses [84], where prolonged temporal context is required. For our task, it was found that shorter window sizes are optimal, but not too short, such as less than 5 seconds. While smaller window sizes could be used for blood pressure, ECG, and EMG, they cannot capture a whole breathing cycle, which is why our model performed poorly at smaller window sizes. The use of a 75% overlap further improves robustness by reducing the step size between windows, smoothing transitions, mitigating the impact of noise, and increasing the effective number of training samples without altering the underlying physiological content.

All models were tested with and without the incorporation of respiratory signals, and including these signals almost always increased the accuracy of the model (Fig. 5). Further validation with the *t*-test solidified the significance of adding respiratory signals on the model's accuracy (Fig. 4). Another key observation is that the single perceptron model, using signal features as input, outperformed state-of-the-art ML and DL models (Fig. 5). This may be due to the small data set, where deeper models (LSTM variants) tended to overfit and performed poorly, while shallow models (decision tree) underfit by failing to capture the non-linear feature relationships of the dataset. As a result, our proposed model struck a good balance between the two and outperformed the other models. Table 7 shows the advantages and limitations of the proposed model against state-of-the-art methods, as well as the use of XAI.

The model used in this study is a single-layer perceptron with 126 input features and no hidden layers, resulting in minimal computational overhead. Each forward-backward pass per sample requires roughly 254 operations. With a batch size of 128, 1908 samples (after windowing), and 700 training epochs, the total number of operations is approximately 3.41×10^8 . The time complexity of our model is therefore $O(N_s \cdot N_E \cdot N_F)$, scaling linearly with the number of samples (N_s), epochs (N_E), and features (N_F), demonstrating the model's efficiency.

Table 7 Comparison of state-of-the-art stroke-risk prediction approaches and the proposed model

Method	Accuracy	Key Advantages and Limitations	XAI
CNN-BiLSTM (EEG) [12]	59.79%	Combines convolutional layers with BiLSTM to capture spatial and temporal features; however, reliance on raw EEG signals increases sensitivity to noise and limits clinical interpretability.	-
LSTM (EMG) [15]	59.80%	Effectively captures fine-grained temporal variations using short windows; very small window sizes are sensitive to noise and may reduce robustness in real-world settings.	-
CNN-LSTM (ECG and PPG) [16]	58.06%	Multimodal input improves predictive performance; increased model complexity leads to higher computational cost and challenges in real-time deployment.	-
Ensemble Echo State Network (EEG features) [19]	66.49%	Efficient temporal modeling using fixed reservoirs and ensemble aggregation; limited interpretability and unclear scalability to heterogeneous multi-modal data.	SHAP for global explanation, and LIME for local explanations. XAI not evaluated.
ML-based models (EEG features) [20]	~80%	Highlights the relevance of low-frequency spectral EEG features for stroke prediction; moderate performance indicates limitations of handcrafted features and potential benefits from more advanced models or multimodal integration.	Eli5 and LIME used. XAI not evaluated.
ML-based models (EMG features) [21]	85.26%	Effectively captures stroke-related gait impairments through EMG spectral features; dependence on gait-specific recordings limits applicability to non-ambulatory or severely impaired patients.	SHAP and LIME used. XAI not evaluated.
ML models (categorical data) [24, 26, 28, 29]	59.49% – 79.87%	Low computational cost and ease of deployment; lacks physiological and temporal information, and class imbalance affects sensitivity to stroke cases.	SHAP and LIME used. XAI not evaluated.
Proposed multi-modal feature-based model	88.23%	Integrates seven physiological signals including respiratory signals (CO ₂ and respiration flow); uses robust statistical and frequency-domain features with a simple model, enabling clinical interpretability and computational efficiency.	SHAP, LIME, and Anchors were each used to provide global and local explanations. XAI results were evaluated using the MEMC and analyzed.

After windowing, we extracted eighteen statistical and frequency-domain features as shown in Section 2.3.2, from each of the seven signals. These features, such as

mean, variance, higher-order moments, and spectral descriptors, capture the tempo-
 ral and spectral properties of each signal while remaining robust to noise and local
 variability. Using the same feature set across signals provides computational efficiency,
 simplifies the pipeline, and allows consistent comparison and fusion of multimodal
 data. Statistical and frequency-domain features have been effective in identifying
 abnormal physiological patterns without requiring explicit waveform segmentation or
 signal-specific morphological analysis [72–74, 85–87].

Using a larger feature set also helped the model in predicting more accurately.
 In the SHAP waterfall plot for the non-stroke subject (Fig. 8), several features with
 high SHAP values increased the probability of a stroke prediction by at least 0.10.
 Nevertheless, the 124 least important features decreased the probability of stroke by
 0.41 when combined. This shows how using a larger feature set helped the model reach
 the correct prediction (non-stroke). The inclusion of frequency-domain features also
 helped in improving the model’s accuracy for the stroke subject (Fig. 8), where the
 ECG2 spectral entropy increased the model’s confidence in predicting the true class.
 This was also seen for the non-stroke subject in the LIME local plot (Fig. 10), where
 the ECG1 FFT energy contributed to the model reaching the correct decision.

The results of the XAI models are in Table 3. The mean feature was the most
 common extracted feature, appearing five times in Table 3. The three most important
 features according to each XAI model are all statistical features showing them as better
 differentiators of the two classes compared to frequency-domain features. The results
 also highlight how all three XAI models identified respiratory signals’ contribution
 towards stroke-risk prediction compared to the rest of the signals used, specifically
 CO₂, where five of the most important features according to all XAI models are derived
 from it. This shows that CO₂ was an important signal in predicting stroke-risk. One
 study [88] discovered that CO₂ is closely related to cerebral oxygen extraction fraction,
 which is an important physiological index of the brain’s oxygen consumption and
 supply, and thus a potential biomarker for stroke. SHAP, which was identified as the
 most reliable XAI model by the MEMC, determined that the CO₂ mean, CO₂ variance,
 and BP mean were the most important features. In other words, the mean values of BP
 and CO₂, and the dispersion of CO₂ values from their average are key determinants in
 predicting whether the subject is at risk of stroke. Moreover, only Anchors identified
 ECG or EMG features as key predictors. The ECG carries extensive information about
 cardiovascular health, which likely explains why the mean was considered important
 [89]. For EMG, the median was highlighted as a significant feature. While similar to
 the mean, the median is more robust to outliers and less sensitive to extreme values
 [90], which is relevant given the inherently noisy nature of EMG signals. This feature’s
 importance is also consistent with the clinical role of EMG in assessing muscle pain
 [91], a condition commonly observed in stroke patients [92].

According to the SHAP plot with feature effects (Fig. 7), a high CO₂ mean is a
 key predictor of stroke-risk. This finding is backed by a study which found that an
 increased level of carbon dioxide, known as hypercarbia, is a stroke-risk factor [93].
 Additionally, the SHAP plot found that a high BP mean is a key predictor for stroke-
 risk. This is also supported by literature as high BP, or hypertension, is a major risk
 factor of stroke [94, 95]. The SHAP plot also found that a low CO₂ variance, or CO₂

signals with little dispersion from the average are a major predictor of stroke-risk. So, a steadier CO₂ signal with less aggressive movement is linked to stroke-risk. This is probably because subjects at risk of stroke had more decrements in respiratory capacity and control than subjects not at risk of stroke [38], which explains why stroke subjects have fewer fluctuations in their CO₂ signals. On the other hand, non-stroke subjects who are still able to use their respiratory capacity to their full capacity are able to attain larger, more rapid changes in their CO₂ signals, leading to a higher CO₂ variance. Nevertheless, these results should be further studied by medical experts to gain deeper insights.

Limitations: One limitation of this study is the lack of publicly available stroke datasets, which restrict the ability to validate the proposed model on external data. Another limitation is the use of a post-hoc explainer, which remains controversial as some researchers argue that explanations should be derived intrinsically rather than after the model is trained [96]. Additionally, while LIME and Anchors are local explainers, they were applied globally by aggregating their explanations across all test cases. Although this was feasible due to the small test sample size, it may become impractical for larger datasets where such aggregation could be time-consuming.

Future Work: Future work can build upon the findings of this study. Notably, the models developed and used in other studies did not perform similarly on this dataset, despite achieving high accuracy on their datasets (>98%). This emphasizes the need to develop models with better generalizability that can reproduce strong performance across different datasets. This is a common challenge in this field. Chen et al. conducted a systematic review of 41 studies that used ML algorithms to build prognostic prediction models for ischemic stroke and 88% of these studies did not validate their models on external datasets [97]. Different and novel validation methods must be used, such as the A-test proposed by Gharehbaghi [98], and the proposed model should also be validated using an external dataset. Furthermore, computation of the 95% confidence interval of accuracies, and the receiver operating characteristic (ROC) curve and area under curve (AUC) values must be performed in future work to obtain a more robust evaluation of the proposed model against other models [99, 100]. Our results show that respiratory signals are important for stroke-risk prediction, and future work should explore whether they can be used on their own without other bio-signals. These results also show that more focus should be placed on CO₂'s role in stroke-risk prediction.

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1279 5 Conclusion

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In this study, we developed a multi-modal stroke-risk prediction model integrating respiratory signals (CO₂ and respiration flow) alongside ECG, EMG, and BP, highlighting the clinical value of underexplored physiological signals. Our results show that respiratory signals, particularly CO₂-derived features, provide unique and critical information for identifying individuals at risk of stroke, complementing traditional cardiovascular indicators, and suggest that future research and clinical investigation of these signals could reveal new predictive biomarkers for stroke and other cardiovascular diseases. This is the first study to demonstrate the predictive importance of CO₂

features in stroke-risk assessment, suggesting their potential as actionable biomarkers. Clinically, incorporating respiratory signals into routine monitoring could enable earlier identification of high-risk patients, support timely interventions, and guide personalized management. Future studies with larger patient cohorts are required to validate these findings and facilitate broader clinical implementation.

6 Declarations

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Data Availability The data that was used for this study is available online: <https://physionet.org/content/cves/1.0.0/data/>

Competing Interests Not applicable.

Ethical Approval Not applicable.

Ethics, Consent to Participate, and Consent to Publish Declarations Not applicable.

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